

Smart Sensor Placement for Device-Free Passive Localization

Thesis Progress Presentation IV

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Agenda

- 1 Paper I: Accepted at IEEE QCNC 2026
- 2 Paper II: The Problem
- 3 Our Approach
- 4 Results
- 5 Next Steps & Timeline



**International Conference on
Quantum
Communications, Networking, and
Computing**

- April 6–8, 2026 Kobe, Japan
- Organised by IEEE
- Track 2: Quantum Computing & Sensing

Accepted Paper

Quantum Optimization for Access Point Selection Under Budget Constraint

Mohamed Khalil Brik, Ahmed Shokry and Moustafa Youssef
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47. Quantum Optimization for Access Point Selection Under Budget Constraint
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The Core Idea: Detecting People Without Devices

Device-Free Passive (DfP) Localization

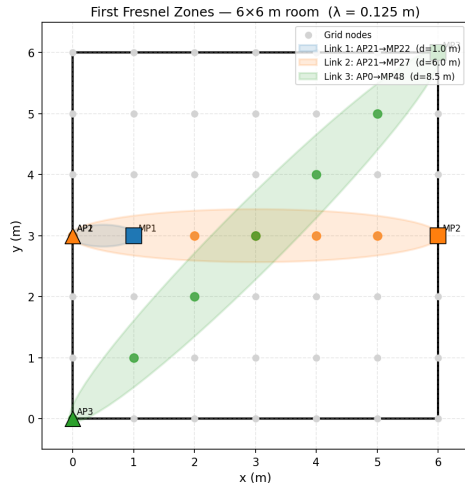
Detect and locate people using **existing WiFi signals**

– no phone, no tag, no active participation required.

How it works:

- WiFi signals travel between routers (APs) and monitors (MPs)
- A person walking through disrupts the signal strength
- The disruption pattern reveals the person's location

Applications: intruder detection, smart buildings



*WiFi links form detection zones.
A person crossing a zone causes*

The Open Problem: Where Should We Place the Sensors?

The challenge from Youssef et al. MobiCom 2007:

“We believe there are classes of configurations – placements of APs and MPs – that have the same performance in capturing changes in the environment.”

In plain terms:

- Sensor placement **strongly affects** detection quality
- Bad placement $\Rightarrow$ large blind spots
- Good placement $\Rightarrow$ nearly full coverage
- But which placement is best?

Our research question:

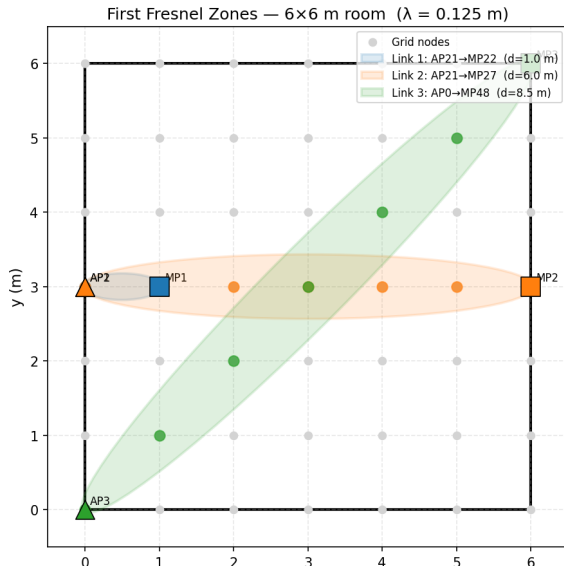
Given a room with **49 possible** sensor locations and a budget of only **3 routers + 3 monitors**, which placement **maximises coverage**?

Number of possible arrangements:

$$\binom{49}{3}^2 \approx 340 \text{ million}$$

Too many to try by hand.

How We Model Coverage



Each WiFi link covers an elliptical area:

- Known as the **first Fresnel zone**
- Any person inside this ellipse disturbs the signal
- Longer links $\Rightarrow$ wider ellipses $\Rightarrow$ more coverage
- Short links cover very little area

Key insight

We want links with **large, non-overlapping** ellipses that together cover the whole room.

Turning Placement into an Optimization Problem

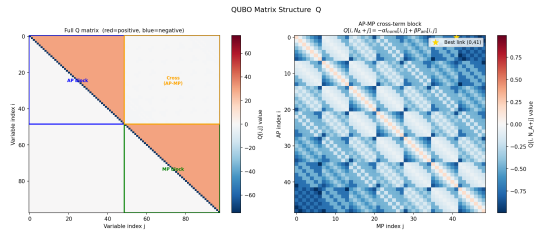
We assign each possible sensor location a binary variable:

- $x = 1$: place a sensor here
- $x = 0$: leave this location empty

The objective captures four goals:

- 1 **Reward** links whose ellipses cover many points
- 2 **Penalise** links that are too short (tiny ellipses)
- 3 **Spread** routers across the room
- 4 **Spread** monitors across the room

Plus a hard constraint: select **exactly** k_A routers and k_M monitors.



The optimization problem is encoded as a matrix of **98 binary variables** (49 router + 49 monitor positions).

Why Quantum Computing?

Classical computers struggle with this type of problem:

- 340 million combinations to evaluate
- Problem grows **exponentially** with room size
- Standard optimizers can get stuck in local minima

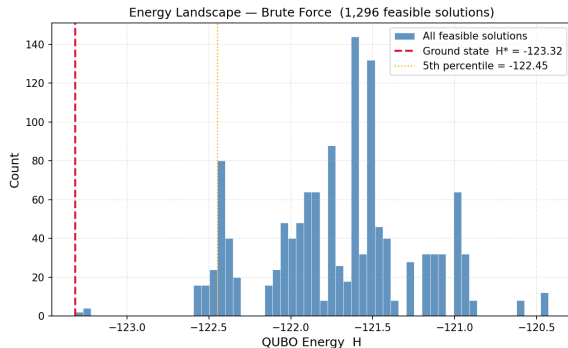
Quantum annealing takes a different approach:

- Explores many solutions **simultaneously**
- Uses quantum effects to escape local minima
- Designed specifically for binary optimization problems

What we do in this work

- 1 Formulate the placement problem as a standard binary optimization (QUBO)
- 2 Validate correctness using brute force on a small instance
- 3 Benchmark classical solvers (SA, Tabu Search)
- 4 Prepare for D-Wave quantum annealer experiments

Validation: Is Our Formulation Correct?



*All 1,296 possible placements ranked by quality.
Our optimizer finds the best one (marked in red).*

Test on a small room (3×3 m):

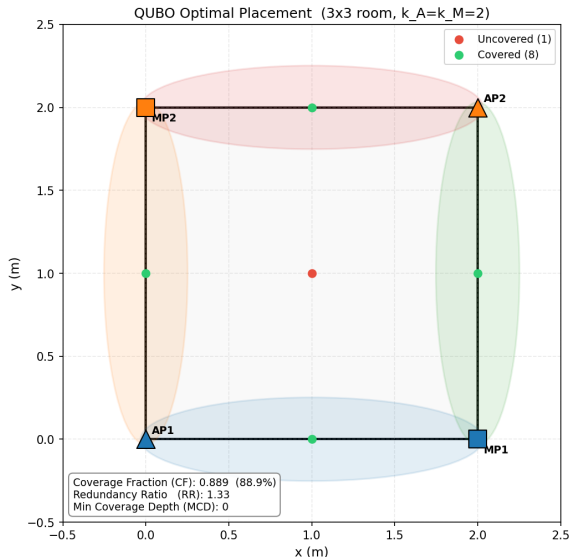
- 9 candidate locations, choose 2 routers + 2 monitors
- = 1,296 possible arrangements – feasible to check all

Metric	Value
Best possible coverage	88.9%
Average random coverage	47.2%
Improvement	1.88×
Optimizer finds best?	Yes

Conclusion

Our optimization correctly identifies the best sensor placement.

What the Optimal Placement Looks Like



Reading the floor plan:

- ● **Green** – monitored area
- ● **Red** – blind spot
- ▲ – router (AP) position
- ■ – monitor (MP) position
- Ellipses – coverage zones of each link

Observation: the optimal solution places sensors at the **corners and edges** of the room – maximising spread and minimising overlap between coverage ellipses.

What Has Been Done and What Comes Next

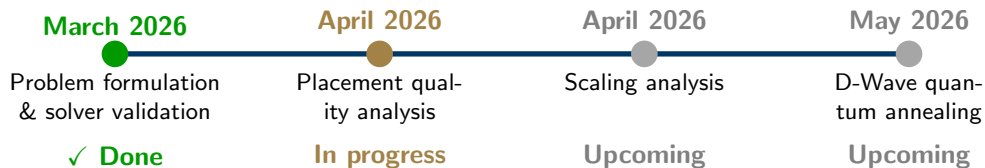
Completed

- ✓ Coverage model built and validated
- ✓ Optimization problem fully formulated
- ✓ Verified: optimizer finds the proven best placement
- ✓ Classical solvers work reliably on full-size room
- ✓ Coverage gain: $1.88\times$ over random placement

Upcoming

- ☐ Compare optimal vs. random vs. uniform grid placement in detail
- ☐ Test how the approach scales to larger rooms
- ☐ Run experiments on a D-Wave quantum annealer
- ☐ Validate on real WiFi sensing datasets
- ☐ Submit second paper

Timeline



Summary

Item	Status
First paper	Accepted at IEEE QCNC 2026, Kobe, Japan
Second paper	Sensor placement optimisation for DfP – in progress
Core idea	Encode placement as a binary optimisation problem
Validation	Confirmed: optimizer finds the provably best placement
Solvers	Always satisfy budget; find exact optimum on small instances
Coverage gain	88.9% vs. 47.2% random $\Rightarrow$ 1.88$\times$ improvement
Next	Scaling study, then D-Wave quantum annealer

Thank You

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